

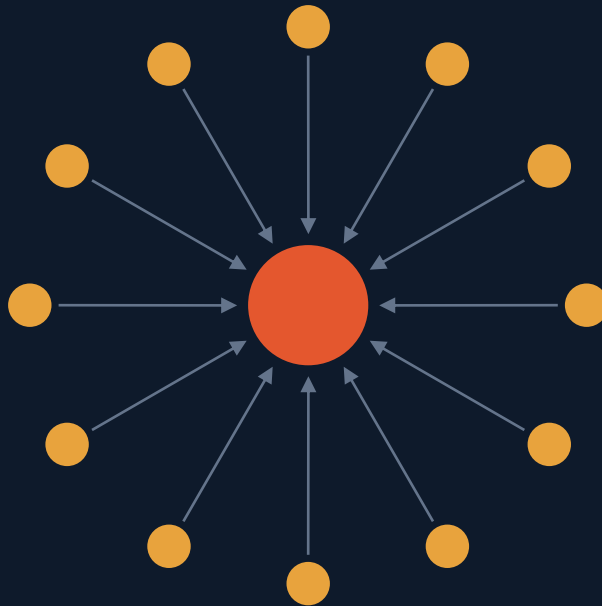
MuleTrace

Risk, Fraud & Regulatory Intelligence Copilot

Finds mule-account networks in bank transactions, explains them with policy evidence, and drafts an audit-ready STR from a plain-English question.

Prototype / MVP Brief

Live demo: muletrace.pages.dev · github.com/SwasthikaDev/SnowflakeCoCoCLI



100 mules → 1 collector

Catch the network, cite the rule, file the STR

THE PROBLEM

Mule networks split lakhs into thousands of small UPI transfers.

Each transfer looks harmless, so rules miss the scheme and analysts trace it by hand for days, then write the STR from scratch.

WHAT WE BUILT

A copilot that sees the network, explains it and drafts the report.

Graph detection of four laundering typologies, evidence and policy clauses on every alert, plain-English Q&A, one-click STR, designed Snowflake-native.

PROOF

4 / 4

laundering schemes caught

100%

precision, 0 false alarms

99.95%

recall on labelled data

Live: muletrace.pages.dev

₹10 lakh, moved as 2,000 harmless ₹500 transfers



Below every threshold

Each ₹500 UPI transfer passes rule-based checks on its own.



Spread across 100 accounts

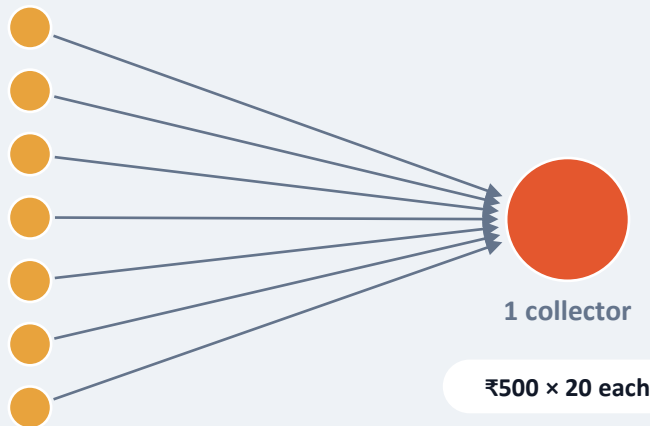
Rented or hijacked mule accounts each look ordinary in isolation.



Visible only as a network

The scheme appears only when you connect who paid whom, when and how much.

100 mule accounts



$$100 \times 20 \times ₹500 = ₹10,00,000$$

Mule accounts are a national-scale banking risk

19 lakh+

mule accounts flagged across India

MHA, February 2025

228 bn

UPI transactions in 2025: the volume that monitoring has to cover

NPCI data, 2025

7 days

to file an STR with FIU-IND once suspicion is established

PML Rules (working days)



Regulatory frame: PMLA 2002, RBI KYC Master Direction and FIU-IND reporting. RBI's innovation hub has launched MuleHunter.AI, so the regulator already treats mule detection as a priority.

Built for the teams who investigate, decide and file



Fraud / AML investigator

Triages alerts and traces money trails

THEY ASK

“Where did the money from this account go in the last 48 hours?”



Compliance officer

Principal Officer who decides on and files STRs

THEY ASK

“Does this breach our structuring policy? Draft the STR.”



Risk & business heads

Own fraud exposure and controls

THEY ASK

“How much money moved through mule-like accounts this month?”

Today the pattern is found by hand, if at all

TODAY

Rules score each transaction alone, so ₹500 transfers pass

Analysts trace hops across screens and spreadsheets for days

Policies and circulars searched by hand in PDFs

Every new question needs SQL from a data team

STR narratives written from scratch



WITH MULETRACE

Graph analysis scores the whole account network

Money trail drawn automatically as an interactive network

Every answer cites the policy clause and the source rows

Ask in plain English; answers respect role permissions

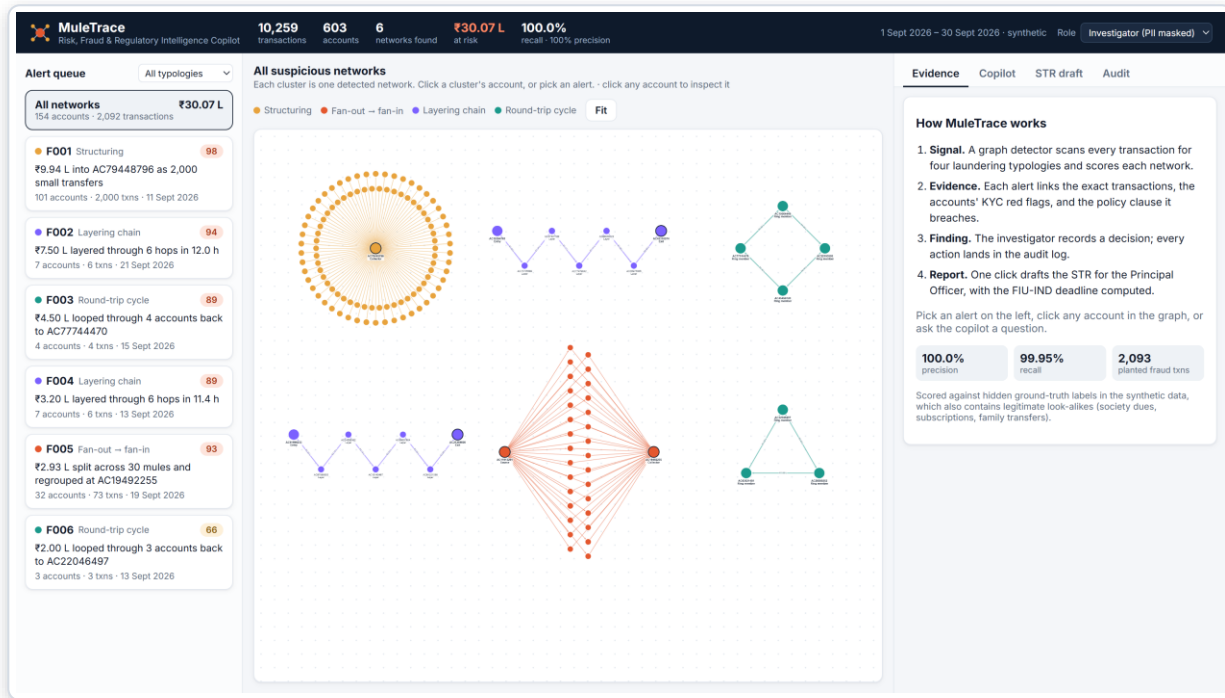
STR draft pre-filled with evidence, ready for review

Only MuleTrace covers signal, evidence and report

	Threshold rules	Black-box ML score	Generic LLM chatbot	MuleTrace
Sees networks across many accounts	X	~	X	✓
Explains why an account is flagged	✓	X	~	✓
Cites the policy clause breached	X	X	~	✓
Answers plain-English questions	X	X	✓	✓
Drafts the STR from the evidence	X	X	~	✓
Keeps data inside the governed platform	✓	~	X	✓

✓ yes ~ partly X no · Typical capabilities of each approach, not specific products.

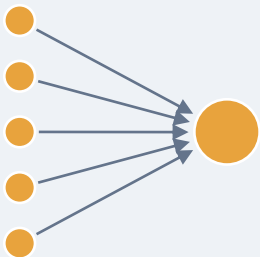
One workspace, from alert to a filed STR



Working app on synthetic data · muletrace.pages.dev

- 1 **Alert queue**
Every laundering network found, ranked by risk.
- 2 **Money trail**
Each network drawn automatically as a graph.
- 3 **Evidence → STR**
Explain, ask, decide and file from one panel.

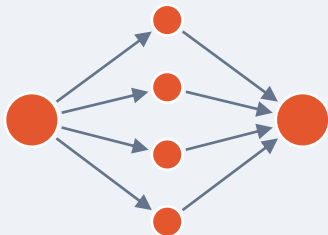
Four laundering typologies, detected as graph patterns



Structuring

Smurfing: many accounts send repeated small amounts to one collector.

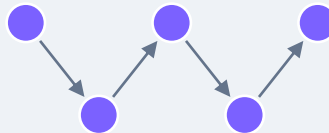
≥10 senders · ≥5 same-amount transfers each



Fan-out → fan-in

Funds split across mules just under limits, then regrouped within hours.

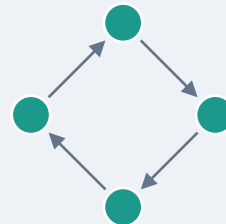
≥8 mules forward ≥70% within 72 h



Layering chain

A large sum hops through 4+ accounts fast, keeping most of its value.

Each hop ≤24 h later, keeps ≥85%

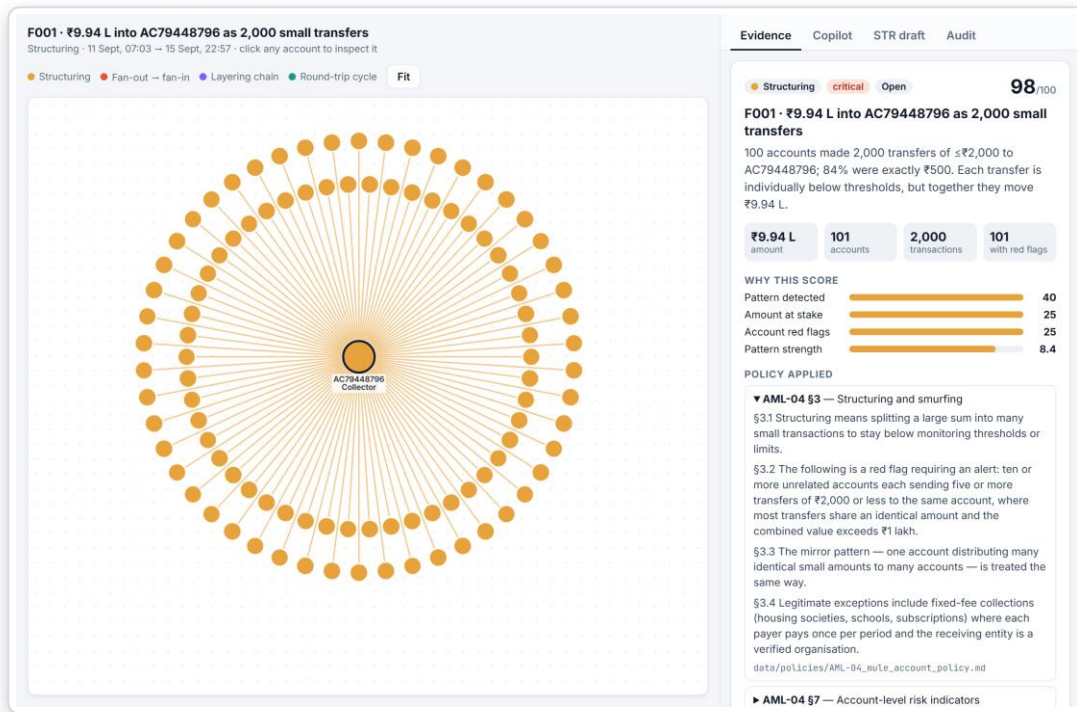


Round-trip cycle

Money returns to where it started, faking turnover or hiding its origin.

Loop closes within 7 days, ≥80% kept

Every alert comes with evidence and a policy clause



1

Explainable score

Four visible parts add up to 98/100. No black box.

2

KYC red flags

New account, minimum KYC, inflow 63× declared income.

3

Policy attached

AML-04 §3.2 quoted beside the alert, with its source document.

4

Decision logged

Escalate, request info or close; every action is audited.

Ask in plain English, get an answer you can defend

Questions run against the transaction graph and the policy text.
Every answer cites its findings, accounts and clauses.

“Which accounts received over ₹5 lakh in sub-₹1,000 transfers?”

→ Finds AC79448796 and quotes AML-04 §3.2

“Where did the money from AC11913291 go within 48 hours?”

→ Traces ₹2.93 L through 30 mules to one collector

“What is our STR filing deadline?”

→ Quotes REG-IN §2: 7 working days

“Who are the 10 riskiest accounts?”

→ Ranked list with roles and red flags

Connected: Snowflake Cortex Agent (Cortex Analyst + Cortex Search). Offline: an in-browser evidence engine keeps the demo working.

Evidence
Copilot
STR draft
Audit

Which accounts received over ₹5 lakh in sub-₹1,000 transfers?

MT
MuleTrace
Evidence engine

1 account received more than ₹5.00 L in transfers below ₹1,000:

- AC79448796** (risk 100) — ₹9.94 L from **2,000** transfers by **100** senders · Collector in **F001** (Structuring)

Policy: this matches AML-04 §3.2 — “ten or more unrelated accounts each sending five or more transfers of ₹2,000 or less to the same account, where most transfers share an identical amount and the combined value exceeds ₹1 lakh.”

Sources
AC79448796
AML-04 §3

Open F001

One click from finding to a filing-ready STR

- ✓ Subjects, roles, KYC data and red flags pre-filled
- ✓ Grounds of suspicion written from the evidence
- ✓ Policy and regulatory basis cited
- ✓ FIU-IND deadline computed: 7 working days
- ✓ Names masked unless you are the Principal Officer
- ✓ Only the Principal Officer can mark it filed
- ✓ Export as Markdown, CSV annex or PDF

Evidence
Copilot
STR draft
Audit

i Markdown
i Annex A (CSV)
Print / PDF
Mark as filed

Suspicious Transaction Report — DRAFT

Status: Draft for Principal Officer review - not yet filed - Confidential — do not disclose to the customer (AML-04 §8.5)

STR reference	STR-20260919-F005
Reporting entity	Demo Bank Ltd (synthetic)
Prepared by	MuleTrace copilot, for the Principal Officer
Draft date	6 Oct 2026
File with FIU-IND by	Thu, 15 Oct, 2026 (7 working days from the suspicion decision, REG-IN §2)

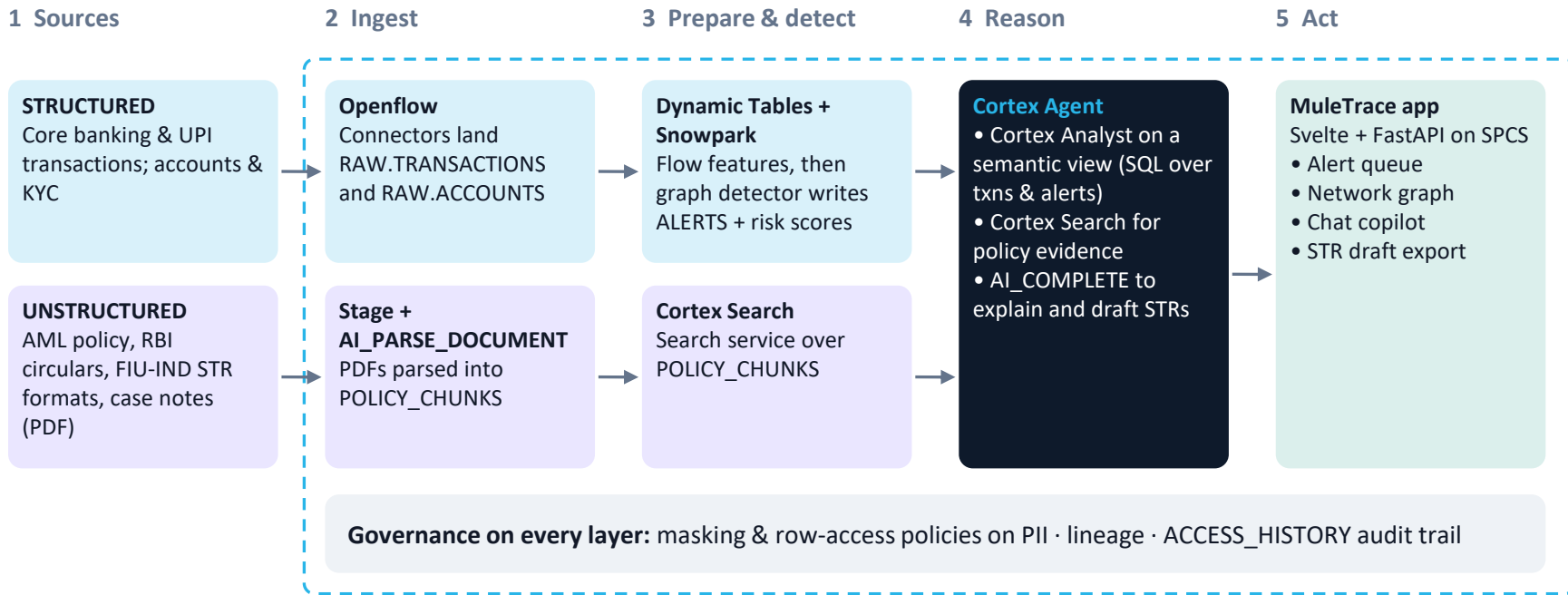
1. Summary

Typology	Period	Amount	Accounts	Transactions	Risk
Fan-out → fan-in	19 Sept, 01:32 – 19 Sept, 12:30	₹2.93 L	32	73	93/100 (critical)

2. Subjects and accounts

Account	Holder	Role	Opened	KYC	Declared income	Risk	Red flags
AC11913291	Rahul Khan	Source	2026-07-30	MIN	₹14,200	93	Account opened 33 days before activity; Minimum-KYC account; Inflow ₹3.00 L is 21× declared monthly income
AC19492255	Ishaan Chopra	Collector	2026-08-02	MIN	₹16,900	97	Account opened 30 days before activity; Minimum-KYC account; Inflow ₹2.90 L is 17× declared monthly

Everything runs inside Snowflake's governed boundary



Dashed line = Snowflake account boundary. Blue = structured data path, violet = unstructured.

Three modular skills: input → processing → output

INPUT

```
> $muletrace-ingest
```

Loads 10,259 transactions, 603 accounts and 17 policy sections into Snowflake and verifies the counts.

PROCESSING

```
> $muletrace-detect
```

Refreshes Dynamic Tables, runs the Snowpark graph detector and ranks 6 laundering networks.

OUTPUT

```
> $muletrace-investigate  
F001
```

Pulls the evidence pack, quotes AML-04 §3.2, logs the decision and drafts the STR with AI_COMPLETE.



Each skill is a markdown file in `.cortex/skills`: reusable by any analyst, versioned in Git, and chained by CoCo CLI into one end-to-end workflow.

12 CoCo CLI skills, each building one module



Ingest & model

openflow

Core-banking feeds into RAW tables

dynamic-tables

Incremental flow features

snowflake-tasks

Scheduled detection runs



Detect & explain

snowpark-python

Graph detector as a stored procedure

ai-functions-pipeline-builder

AI_COMPLETE alert explanations

document-intelligence

Parse policy & circular PDFs



Ask & answer

search-optimization

Cortex Search over policy text

agent-studio

Cortex Agent + semantic view

skill-development

Custom /investigate skill for analysts



Govern & deliver

data-governance

Mask PII, row-level access

lineage

Evidence trail for auditors

deploy-to-spcs

Ship the Svelte investigator app

Signal → evidence → finding → report



Modules talk only through governed tables. A new typology, another UI or a different detector plugs in without touching the rest, and every step is in Snowflake lineage.

Every planted scheme caught, no false alarms

4 / 4

schemes caught

structuring, fan-out → fan-in, layering, round-trip

100%

precision

2,092 of 2,092 flagged transactions were fraud

99.95%

recall

2,092 of 2,093 planted fraud transactions found

0

look-alikes flagged

society dues, subscriptions and family transfers left alone

₹30.07 L

laundered value surfaced

6 networks, 154 accounts

0.1 s

detection time

10,259 transactions on a laptop

A synthetic month of banking data with hidden ground-truth labels the detector never sees. Reproduce with: `python backend/pipeline.py`

From days of manual tracing to minutes

TARGET

Days → < 5 min

From alert to a full evidence pack for a ring of 100 accounts

TARGET

< 1 min

To a first STR draft, so analysts review instead of write

DESIGN GOAL

100%

Of answers cite source transactions and a policy clause

MEASURED IN DEMO

100% / 99.95%

Precision / recall against ground-truth labels planted in the synthetic data

Targets are goals for the prototype, to be validated against a bank's own baselines in a pilot.

Built for UPI-scale volume, and for more than fraud

Scales with Snowflake



Elastic compute

Warehouses scale up for month-end scans and down when idle.



Incremental by design

Dynamic Tables process only new transactions.



Bounded graph search

Time-windowed and partitioned by bank, branch or day.

Extends beyond the demo



More typologies

Dormant-account revival, rapid KYC changes, cash-in to UPI-out.



Cross-bank intelligence

Share mule signals via Data Clean Rooms without sharing PII.



Wider risk copilot

Same agent for credit, liquidity and Basel reporting questions.

From working MVP to a bank pilot

NOW

MVP · Oct 2026

- Live investigator app
- Graph detector with measured accuracy
- Copilot, STR drafts, audit log
- Snowflake SQL + 3 CoCo skills

NEXT

Grand Finale · 27–30 Oct

- Full workflow running in Snowflake
- Cortex Agent answering in the live app
- Hourly detection task + alerts
- PDF policies via AI_PARSE_DOCUMENT

THEN

90-day bank pilot

- Anonymised production data
- Thresholds tuned with investigators
- Time-to-STR measured against baseline
- Cross-bank signals via clean rooms

How MuleTrace scores on the rubric



Real-world relevance 30% of score

- Built for Indian banking reality: UPI-scale mule networks, PMLA, FIU-IND STRs
- Follows how AML teams actually work: alert, evidence, decision, report
- Respects PII masking, tipping-off and record-keeping rules



Technical execution 40% of score

- Time-ordered graph detection: 100% precision, 99.95% recall
- The same detector runs locally and as a Snowpark procedure
- Cortex Agent, Cortex Search, semantic view, Dynamic Tables, AI_COMPLETE



Solution completeness 30% of score

- Covers the full flow: signal, evidence, documented finding, report
- Joins transactions and KYC data with policy and filing text
- Working app, open code, reproducible data and audit trail

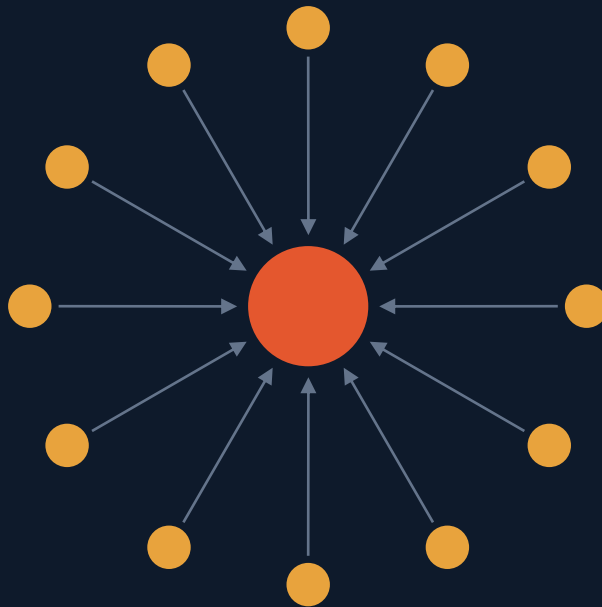
MVP DEMO SCOPE

In the demo

- ✓ A month of synthetic banking data with 4 hidden schemes
- ✓ Graph detection and a risk score for every account
- ✓ Interactive network diagram for every case
- ✓ Plain-English Q&A with cited policy clauses
- ✓ One-click STR draft with evidence attached

Live demo muletrace.pages.dev

Code github.com/SwasthikaDev/SnowflakeCoCoCLI



MuleTrace: follow the money, file with confidence.